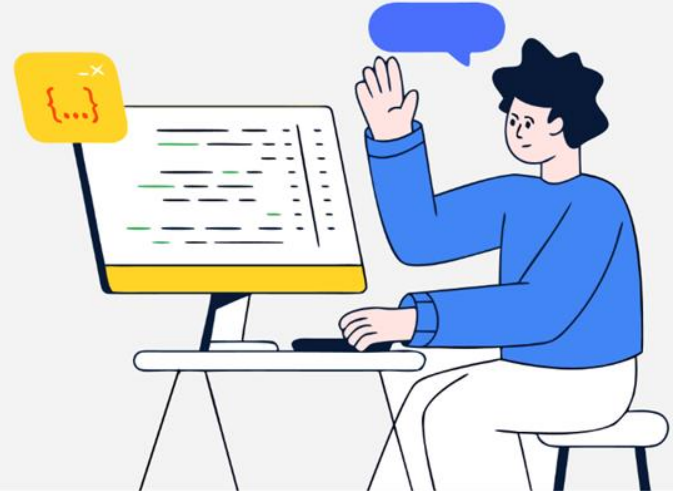




Solution Challenge



Team Details

- Team name:** Team X .
- Team leader name:** Mrinal Roy .
- Problem Statement:** Digital Asset Protection Protecting the Integrity of Digital Sports Media

Brief about our solution

ALETHEIA is a cutting-edge web application designed to restore trust in digital media. By combining industry-leading forensic detection with advanced linguistic explanation, it provides a comprehensive solution for unmasking AI-generated content.

Core Capabilities

Aletheia doesn't just give you a "yes" or "no." It provides a deep-dive forensic report by leveraging a unique dual-API architecture.

- **Detection Accuracy:** Utilizing **Sightengine's** specialized forensic API, the platform delivers a precise probability score (0–100%) for images, videos, and audio. It is tuned to recognize the unique pixel and frequency patterns of major generators with **99% accuracy**.
- **Intelligent Breakdown:** Powered by **Google Gemini**, the engine translates technical data into a human-readable analysis. It identifies specific "artifact indicators" (like unnatural textures or warped geometries) and "authentic signals" to explain *why* a file was flagged.
- **Generator Identification:** The system can often assume the specific model used to create the media, such as **Midjourney, DALL-E 3, Stable Diffusion, or Sora**.

Opportunities

a. How different is it from any of the other existing ideas?

- Multi-modal support (image + video + audio + social posts in one place)
- Dual-AI pipeline (detection + explanation layer)
- Interpretability focus (breaking down why something is AI-generated)

b. How will it be able to solve the problem?

- A 78% AI score alone is confusing
- The explanation layer translates it into human reasoning (artifacts, inconsistencies, etc.) Handling real-world messy inputs
- Direct CDN links, compressed files, reposted media
- This is where many academic detectors fail Reducing user effort
- One platform instead of using separate tools for images, videos, and text Improving trust
- By showing technical + visual evidence, users understand and verify the result

c. USP of the proposed solution

- Not just what is AI, but why Two-layer intelligence system
- Detection engine + reasoning engine (powered by Google Gemini) Cross-platform compatibility
- Works on real social media content, not lab data Generator identification
- Guessing tools like Midjourney or Stable Diffusion adds practical insight Real-world usability
- Supports multiple formats + direct links (huge for journalists, researchers, general users)

List of features offered by the solution

- Detects whether an image, video, audio or social media post is ai generated or real using Sightengine's Api with 99% accuracy. Returns a precise probability score from 0-100%.
- Powered by Google Gemini, generates a human-readable breakdown including artifact indicators, authentic signals, technical details, and a likely AI generator guess (Midjourney, DALL-E, Stable Diffusion, etc.).
- Supports multiple file formats like .MP3, . MP4, .jpeg, .jpg and direct CDN image URLs from any platform.
- Uses two AI APIs together — one specialized for detection, one for explanation.
- Not just a percentage — gives visual + textual + technical breakdown.
- Works on real-world images from the web, not just test datasets.

Process flow diagram or Use-case diagram



PROCESS FLOW - v1.0

SLIDE 02

Process Flow Diagram

End-to-end pipeline from user input through dual-API analysis to forensic verdict



REQUEST - UPLOAD

```
POST /api/analyze/upload
Content-Type: multipart/form-data
Field: image (buffer) - Max 20MB
Types: JPEG / PNG / GIF / WEBP
```

REQUEST - URL

```
POST /api/analyze/url
Content-Type: application/json
Body: { "url": "https://..." }
Timeout: 10s - fetched server-side
```

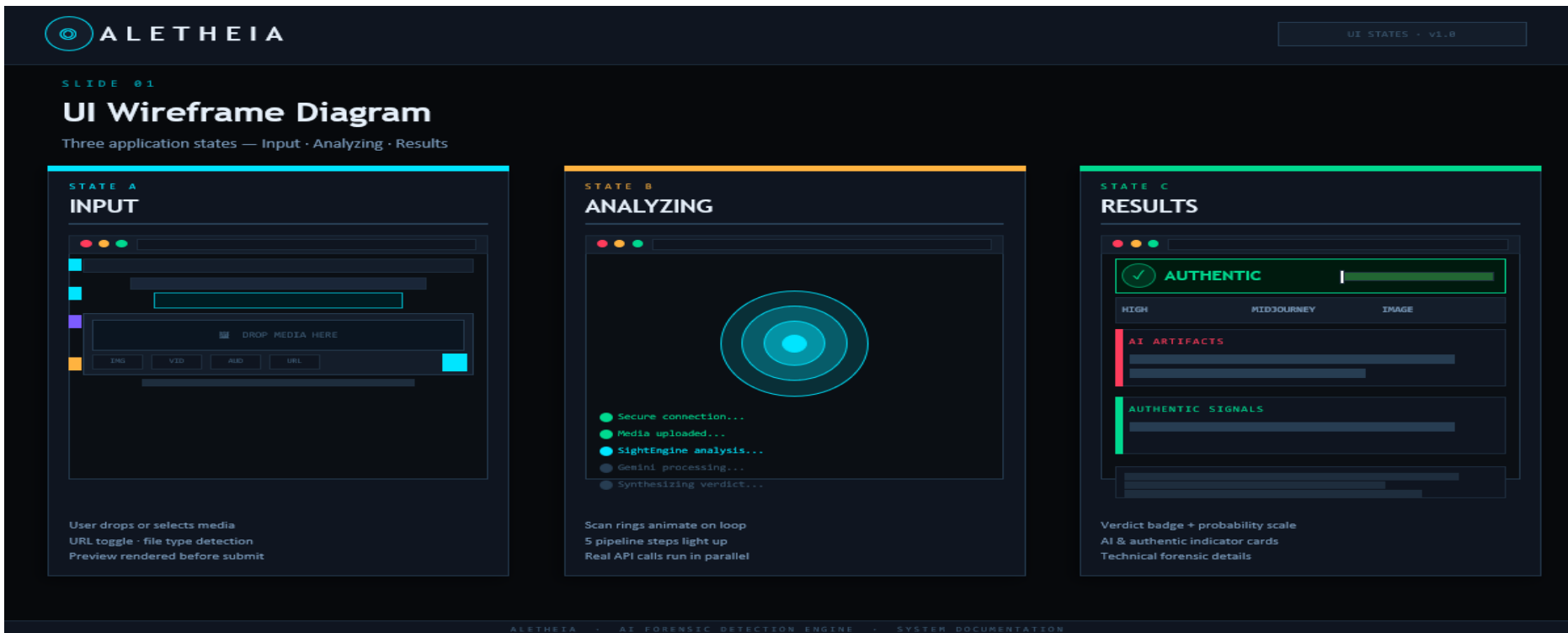
RESPONSE SHAPE

```
verdict: "ai" | "real" | "uncertain"
aiProbability: 0-100
confidence: "High"|"Medium"|"Low"
aiIndicators[] - realIndicators[]
```

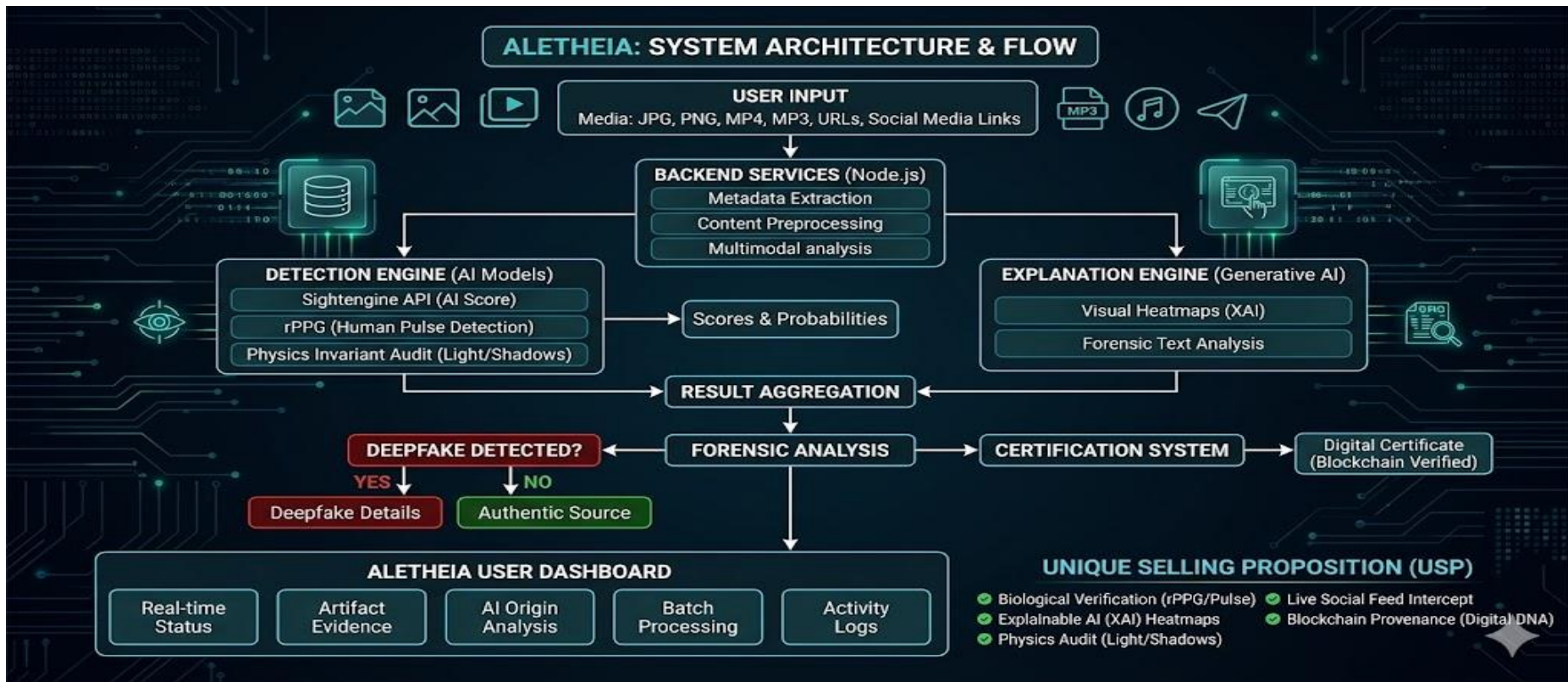
MERGE LOGIC:

If Gemini responds → weighted blend: 70% Gemini + 30% SightEngine. If Gemini fails → SightEngine only (confidence: Medium). If both fail → uncertain (confidence: Low).

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies to be used in the solution

Frontend



Html



Css



Js

Ai APIs



Sightengine



Google gemini

Backend



Node js



Express

Estimated implementation cost

- **Current Cost (Prototype Stage):**
₹0 (Built using free-tier services and credits)
- **Current Limitation:**
System supports ~10,000 tokens/day due to API free-tier restrictions

Estimated Cost (Scaling Phase)-

AI Processing (Google Gemini API): \$10–30/month (depending on usage)

Forensic Detection API (Sightengine): \$20–50/month (based on media analysis volume)

Cloud Hosting (Google Cloud / Firebase): \$5–20/month

Total Estimated Cost: \$35–100/month (₹3,000 – ₹8,000 approx.)

Scalability Plan -

- Pay-as-you-go model ensures cost grows only with usage
- Can optimize costs using:
 - Caching frequent results .
 - Efficient token usage .
 - Batch processing .

“The solution is currently cost-free at prototype level and can scale affordably with demand.”

Snapshots of the MVP



Future Development

◆ **Expand to Full Multi-Modal Detection**

Support advanced analysis for images, videos, and audio using unified forensic models.

◆ **Real-Time Verification Tools**

Develop a browser extension and API for instant authenticity checks on social media and web platforms.

◆ **AI Model Signature Database**

Continuously update detection for emerging generators like Midjourney, DALL·E, Stable Diffusion, and Sora.

◆ **Explainable AI Enhancements**

Provide visual highlights (artifact regions, waveform anomalies) along with human-readable explanations.

◆ **Enterprise & Media Integration**

Offer scalable APIs for news agencies, fact-checkers, and platforms to automate fake content detection.

◆ **Blockchain-Based Authenticity**

Enable tamper-proof verification and digital certificates for verified media.

◆ **Continuous Learning System**

Improve detection accuracy using real-world data and user feedback loops.

“ALETHEIA aims to set the global benchmark for digital media authenticity in the age of AI.”

Provide links to your:

1. GitHub Public Repository - <https://github.com/Mrinalray/Aletheia>
2. Demo Video Link (3 Minutes)-
https://drive.google.com/file/d/1MeuyqtFlmT_Z5is94XPKOBRXvCtlz0ns/view?usp=drivesdk
3. MVP Link - <https://aletheia-mvp.netlify.app>
4. Working Prototype Link - <https://aletheia-deepfake-tool.netlify.app/>

Team-X



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Lead



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Solution Challenge



Thank You